

Trump Rally Transcripts — NLP Analysis

This notebook runs a multi-method NLP pipeline over 103 Trump 2020 campaign rally transcripts (`Trump_Rallies_Dataset.csv`). Broadly, it moves through four kinds of analysis:

1. **Named entity tracking** — who gets mentioned, and how often (Sections 3–4)
2. **Corpus-level structure** — how similar different rallies are to each other, and what audiences react to (Sections 5–7)
3. **Sentiment feature analysis** — which words drive positive vs. negative sentiment in a separately labeled phrase set (Sections 8–9)
4. **Topic modeling** — what Trump actually says about Joe Biden specifically, via two different topic modeling approaches (Sections 10–12)

Section titles and explanations below were added for readability; the underlying code is unchanged from the original notebook. A couple of small bugs are flagged inline where they'd stop a cell from running top-to-bottom.

1. Setup & Data Loading

Loads the rally transcript dataset and does a first pass inspection (`.head()`, `.shape()`) to confirm the data loaded correctly — 103 rows, one row per rally, with `Place`, `Date`, and the full `Transcript` text.

```
In [ ]: import pandas as pd
```

```
In [ ]: df = pd.read_csv(r"/content/Trump_Rallies_Dataset.csv")
```

```
In [ ]: df.head()
```

	Unnamed: 0	Place	Date	Transcript
0	0	Speech: Donald Trump Holds a Campaign Rally in...	October 12 2020	Hello, everybody. Hello, Orlando. Hello, Sanfo...
1	1	Speech: Donald Trump Holds a Campaign Rally in...	September 30 2020	Hello, everybody. Hello, Duluth. Hello, Duluth...
2	2	Speech: Donald Trump Holds a Campaign Rally in...	September 26 2020	Oh, thank you. Well, we won Pennsylvania last ...
3	3	Speech: Donald Trump Holds a Campaign Rally in...	September 25 2020	Thank you very much. Hello, Newport News. I kn...
4	4	Speech: Donald Trump Holds a Campaign Rally in...	September 24 2020	[This is a machine transcript written automati...

```
In [ ]: df.shape
```

```
(103, 4)
```

2. Inspecting a Sample Transcript

Pulls out a single transcript (row 92) in full, as a sanity check before running any NLP on the whole corpus.

```
In [ ]: transcript_92 = df.loc[92, 'Transcript']
```

```
# Display the isolated transcript
print("Transcript 92:")
print(transcript_92)
```

Transcript 92:

Thank you. Thank you. [Applause, Crowd chants USA] Thank you, everybody. Thank you. And if you think this crowd is big, you should see right now what's outside. Congratulations. [Applause] Hello, Pensacola. What a place. Right here on the Emerald Coast, right? The Emerald Coast in the great state of Florida where we had a tremendous victory, didn't we? Come on. [Applause] We are joined by many, many people in the military and great American patriots. Thank you very much. [Applause] When you remember the campaign, I said, let me begin by wishing each and every one of you a very Merry Christmas. Right? [Applause] And I can think of no better Christmas present for the American people than giving you a massive tax cut. That is what's happening. [Applause] And your Senate and your House hopefully very shortly will have those approvals. Those heads will go up and you will be paying a lot less tax. So, congratulations. I hope. They got to vote and I think they will. I can't wait to sign that tax cut great for so many. But great for the middle class. We're going to put it into law. We're going to cap off an incredible first year in office. There has been no first year like this. Or at least just about -- no, I think the answer is no. But I have to be very accurate because of the fake news back there. [Applause, Crowd Chants Trump] Thank you. In our campaign for the presidency, we defied the pundits, the politicians, and the special interests because ours was a movement of the people and it really was. It was a movement of you. It was a movement of the people. It was a movement to make America great again. It's happening. It's happening. And by the way, how are your 401(k)s doing? Not too bad, right? [Applause] Over the weekend, I was in New York City. And I love to say hello to our folks in uniform whether it's the military, whether it's law enforcement. And we had people backstage and I always want to take my time. And a man comes up. He was a police officer in New York City. The first time anyone said this to me. He said, Mr. President, I want to thank you. My family thinks I'm a financial genius. My 401(k) is up 39 percent in nine months. [Applause] I think it's going to be very hard for somebody to beat us in a few years. Can you imagine now? We are only talking about a few years. All you have to say is, with us it goes up with them, it goes down and that's the end of the election. Right? [Applause] Everyone is doing well and it's going to be a lot better. We are going to do better and better. We are going to make our country great again. And we are looking at our country now as America first. America first. [Crowd Chants USA] So tonight we are going to speak straight to the American people and cut right through the fake news media. Right through. [Applause] Right through. We're going to speak the plain truth and really the truth that you just want to hear. You have to hear. They don't want to hear. But, and by the way, did you see all the corrections the media has been making? They're saying sorry, we made -- they have been doing that all year. They never apologize. Maybe that comes with being the President. I don't know. They have been apologizing left and right. They took this fraudster from ABC. They suspended him for a month. They should have fired him for what he wrote. [Applause] He drove the stock market down 350 points in minutes which, by the way, tells you they really like me, right? When you think of it. And you know what he cost people? And I said to everybody, get yourself a lawyer and sue ABC News. Sue them. [Applause] And then

CNN apologized just a little while ago. They apologized, oh thank you CNN, thank you so much. You should have been apologizing for the last two years. [Applause] True. What a group. What a group. But since the election and then I am going to argue with this. We created 2.2 million new jobs. Factories are coming back to our country. You know we have factories pouring back into our country? Did you ever think you would hear that? I used to tell you that's going to happen. See now there is consequences when companies close up their factories and move to another country, make a product and sell it right through. They have consequences now. And by the way those consequences are getting very steep. And they are going up. There is no more of that stuff. There is no more. In November alone, you saw this today we added 228,000 jobs. I watched these anchors they were devastated. Oh, what's going on? And honestly we have a long way to go. We are going to do much better than this and I have been saying it. Including another 33,000 new jobs in manufacturing. You hear that, right? Hispanic unemployment is the lowest ever recorded. Any Hispanics here? Any Hispanics? Any Hispanics? You guys were great. Remember? They weren't going to vote for me because, I'm going to build the wall but they want the wall, too. They want security. [Applause, Crowd chants Build That Wall] Oh, we're going to have the wall. Don't worry. They want security. How about the Cuban population in Miami? Well, we have Cubans here tonight, good. Was the number 86 percent in favor of Trump? A tremendous percentage. Total unemployment is now at a 17-year low. Think of that. Not bad. Not bad after 10 months. It's only going to get better. Listen to this one. This is to me maybe the most important because it's all psychological to a large extent and that's what creates great news. Consumer confidence is at 17-year high. Think of that. It was not like that in your last administration. Economic growth last quarter surged to 3.3 percent. You know it. You know where it was, right? When we started, you know where it was? Bingo. 3.3 percent and it's going a lot higher. You know, I used to hear you will never hit three. You will never hit 2.5. You figure maybe 2.3, 2.4. If we didn't have the hurricanes last quarter, we would have hit four percent and everybody back there would have said that's impossible. We had a little -- couple of hurricanes. You heard about it, right? You heard about it? The stock market is hitting an all-time high record for another -- and think of this. Eighty six times since Election Day, 86 times. [Applause] It hit a record. Eighty six times. And we're going to keep it going. And then you look at all of the money you folks are making. Oh, I wish I could take 10 or 15 percent. But I think you are not going to do that. But it's getting bigger and bigger. Your pensions are getting bigger and bigger. Your pension funds, all of those things are happening. And wait until you see what we're doing. Tax cuts. But wait until you see what we are doing with healthcare. Wait until you see. You're going to be happy. Not the ObamaCare nightmare. Not ObamaCare nightmare. It's not that nightmare where your premiums are a disaster, right? They are a disaster. Now we have it going you watch we have some very good things happening with respect to healthcare. And the taxes are a part of it. And you will see what happens right after taxes, you will going to see what happens. Because with taxes something great is going to get announced. We're getting rid of the individual mandate. That individual mandate. [Applause] Where you pay a lot of money for the privilege of not having to have insurance or healthcare. So you pay for the privilege of not getting taken care of. Isn't that a wonderful thing? And we're going to repeal it. [Applause] Since the election, we have created more than \$5 trillion in new economic wealth just in the stock market alone. We are not including real estate and other values, \$5 trillion. And we are finally rebuilding our country, defending our citizens and fighting for our great American workers. [Applause] We're putting America first by renegotiating trade deals. We are renegotiating NAFTA. We are renegotiating our horrendous deal with South Korea. I kept you out of the TPP which would have been a total disaster for you and for everybody else. And we're making individual deals with these countries. You know, when you get tied into 12 countries or 22 countries or 18 countries, you are stuck. It's a spider web. It's like this. If you have 16 good ones and two bad ones as an example, you can't do it. You can't get out. When you have a deal one-on-one. Nice simple deal. We have a deal with Japan or we have a deal with South Korea or we have a deal with whoever it may be if they don't treat us right, we send them what's called a letter of termination. [Applause] These people always say, oh TPP. You have all these countries tied. And you can't terminate. You are tied into all the good ones. [Audience Member calls out] Thank you very much. I appreciate it. So, you can't do what you have to do. So you send out a letter of termination. You give them a 60-day notice or a 30-day notice or even a six month notice. They then come back because we're the big piggy bank that everybody likes robbing. The whole world robs it. So they then come back and they say, can we negotiate? Can we make a better deal than we had in the first place, right? But you can't do that when you have many countries tied in. You can't -- I call it a spider's web. There is a lot of Washington lobbyists, bureaucrats, politicians who don't want to see things changed. They have made fortune. Areas around Washington, D.C. are the richest areas in this country. You saw it the other night. A list came out. We have five areas that are among the richest areas in the country. That's because everybody is making a fortune and people outside of that area are paying for the money they are making we're not going to have it. They like things the way things used to be. They don't like it so much now. Although, I must tell you they are doing well also. They're doing well also. Their stocks are going up. I don't know if there is anything I can do to prevent it for those people. But they call themselves the resistance. Do you ever see these signs resist. Resist. [Applause] I love these guys. Look at these guys, blacks for Trump. I love you. I love you. I love you. [Applause] By the way, now that you bring it up black home ownership just hit the highest level it's ever been in the history of our country. Congratulations. [Applause] Oh, these resisters resist. Hillary resisted and do you know what happened? She lost the election in a landslide. [Applause] But you know what they are really resisting? They are resisting the will of the American people. That's what they are resisting. [Crowd Chants Lock Her Up] Look, it's being proven we have a rigged system. It doesn't happen so easy. But this system, there will be a lot of changes. This is a rigged - - this is a rigged system. This is a sick system from the inside. And, you know, there is no country like our country but we have a lot of sickness in some of our institutions and we're working very hard. We have a lot of them straightened out. But we do have, we really do. We have a rigged system in this country that we have to change it. Terrible. Terrible. They are resisting progress. They're resisting change. Because the only thing they really care about is protecting what they have been able to do, which is really control the country and not to your benefit. By the way, wages starting to go up. First time in 20 years starting to go up. That's all going to happen. That's all going to happen. [Applause] That's all going to happen. That's just a part of what's going on. But I am going to do something better than that. We are going to have choice. Now, we have choice in education now. We are starting that. Right? And I thought of it the other day. We're going to have choice in jobs so that you don't have one job and hang on even though you don't like it. You can have choice where can you take five jobs, six jobs, seven jobs, pick the one you like at the most money. That's what is starting to happen because the jobs are coming in. That's what is starting to happen. Thanks to General Mattis and the military leaders and the allies. Mad Dog. Mad Dog Mattis. [Applause] Mad Dog Mattis. Thanks to Mad Dog Mattis that we have great military leaders. ISIS is being dealt one brutal defeat after another. You see it. Not only are we defeating these killers, these savage killers, horrible, horrible, you don't even want to say people. These are savage killers. Over there but we sure as hell don't want them to come over here. I'm sorry. I'm sorry. [Applause] You know, they come back to some countries and they come in. We are making it a very difficult process. We had such weakness. They go out, kill people. And then they come back and they go back home to mom and dad. Okay? Before they went nutso and start over here, we are making it very difficult. You see what's being happens. We are watching every single one. We have thousands of people right now under surveillance. Surveillance. That sounds familiar. That sounds familiar. Remember what I suggested something like that everyone said Trump, why does he say? Well, it turns out I was right about that one, wasn't I? [Applause] America is being respected again abroad and we are taking care of our citizens at home. And we're going to have safety. And we have got a lot more now. We are getting rid of the M.S. 13 animals. [Applause] We're going to have safety again. Because America is more than just a place on a map. America is a nation, America is a family. America is ours to love and to cherish and to protect and to take care of and that's what we're doing, we're going to take care of this country for our children and our grandchildren and our great grandchildren. And we're not going to let people come in to our country who are going to destroy our country. [Crowd Chants USA] And that may or may not be politically correct. But honestly, I don't care. We have to do the right thing. We have to do the right thing. [Applause] Our agenda is pro-family, pro-police, pro-worker, and 100 percent pro- American. [Applause] That's why I recently withdrew the United States from the United Nations plan for global governance of immigration and refugee policy. No borders. They have a plan, no borders. I heard about it recently. I said, what are you talking about, no borders? No borders. No borders. Everybody can come in. If you don't mind, I rejected that plan. Is that okay? [Applause] I told them not only do we want no borders. We want the strongest borders you have ever seen. We're going to have such borders. We'll have borders on top of borders. American is a sovereign country. We set our immigration rules. We don't listen to foreign bureaucrats. We don't listen to other countries telling us how we should be running our immigration. Thank you very much. [Applause] We're also getting smart and tough on trade. I have been talking about this for a long time. It's one of the reasons I'm here. One of the reasons I'm president. We are renegotiating the disastrous trade agreements of NAFTA, of South Korea,

of so many others. We are making great deals. We're going to hopefully keep NAFTA but there is a chance we won't. And that's okay. That's okay. How many factories have left for Mexico? How many factories? [Crowd responds Too Many] So we right now have a trade deficit with Mexico of \$71 billion and that doesn't include all the drugs that pour in over the border because we don't have a wall, et cetera. Okay? \$71 billion. Seventy one billion. So, we have, think of it, we have a trade deficit with Mexico of \$71 billion. We have a pretty good trade deficit with Canada. They were saying, we have a surplus with Canada. I said, no, and I like the prime minister very much. Prime Minister Trudeau, nice guy. Good guy. No, I like him. But we had a meeting. He said no, no. You have a trade surplus. I said no, we don't. He said no, no, you have a trade surplus. I said, Mr. Prime Minister, we do not. He said how, do you know? I said, because we don't have a surplus with anybody. We have the worst deals. He said I'm telling you that Canada has a deficit with the United States. I told my people, in front of a lot of people. I said, go out and check. And he was right except he forgot two categories, lumber, timber and energy. Other than that, he was right. When you add them all together, we actually have a \$71 billion deficit with Canada. Right? So, he forgot a couple of categories that he didn't want to mention. I heard it with other countries too. I just got back from Asia. I brought back \$300 billion worth of deals. Worth of deals. [Applause] And that number is going to go a lot higher. And these are numbers that are going to produce a lot of jobs in our country. And that's what we have to be doing. We are going to end the theft of American intellectual property, crack down on China's trade abuses, and confront countries that cheat, of which many of them are in that category sadly. We will buy back our country. We will take back our country. We will make those great trade deals and we will have unbelievable amounts of jobs created and factories created. [Applause] And we will do what other countries do. We will start buying American. We want to buy our product. And we will start hiring American. And we will say American and be proud of it and that's what's happening more and more. That's why I see all those red hats and those white hats. I love those hats. [Applause] That is why one of my very first acts as president was to withdraw the United States from the job-killing Trans-Pacific Partnership. Had to do it. We have also cancelled the one job killing regulation that you see after another. One after another, beating the mark of any previous president. Now, we have -- there was a great article in the "Wall Street Journal" the other day You know who was right up there? Honest Abe Lincoln. Can you believe it? [Applause] He was a regulation cutter. Can you believe it? Abe Lincoln was a regulation cutter. Who would've known that? I said, you mean I beat Abraham Lincoln? That's pretty good for 10 months. I don't know if regulations sound so glamorous but tell you it's very important. But we have a lot of others. We have statutory requirements where you have to give a 90-day notice and then another 90-day notice, then you have a cooling-off period then you have to give a 30-day notice. They're all in the works. And you going to see a lot more cutting of regulations. But we've done more than any other president and it's not even close - and we haven't even started. And as promised - and I used to come out here - I've been here three times - I love it. I just keep coming back. I just keep coming back. [Applause] Oh boy, oh boy. I'll tell you. When we were doing well during that beautiful election night - when we were doing well all over Florida I said, was the Pensacola area included yet? And they said, no. I said, guess what, we just won that election. [Applause] Right. And as promised, remember? I announced that I would withdraw America - the United States - from the horrible, for us - good for other countries - it's great for other countries - from the horrible Paris Climate Accord. [Applause] Cost us a fortune. So, China doesn't start until around 2030, I think. Russia doesn't have to go back to like a recent date. Go back to somewhere in the 1990s which was a high-pollution time. Other countries we end up giving money to. This would've been one of the great catastrophes - and I could come back into the deal at a much better price. They'd love me back in. I could come back into the Paris Accord - and I like the people. I really like Emmanuel. I like so many of the people involved but I could come back - it's like - I don't get it. I don't see it. I want clean air. I want crystal clear and clean water and I want things that they won't have - and I don't want to pay the kind of numbers that you're talking about. If we stayed with those numbers, we would have had to close factories and businesses in order to qualify by 2025. Can you imagine, ladies and gentlemen, we're violating the accord. In the meantime, some of those countries are spewing stuff out that you wouldn't believe - and we would be closing up. Now we could maybe change the deal but then we get sued by all the groups that sue. You know in China - they don't see China; nobody sues. With us, they sue. I approved - immediately thereafter we took office the Keystone XL and Dakota Access Pipeline; right? 48,000 jobs. We've lifted the restrictions on American energy including shale, oil, natural gas and clean beautiful coal of which we have 1000 years of supply and we're putting miners back to work. Did you see West Virginia? I love West Virginia. West Virginia. I love West Virginia. So, West Virginia people were saying - trouble - coal will never come back. In the meantime, they're buying it in Vietnam from us. The president of Vietnam said, that is the finest coal we've ever purchased. They're buying it from us. So, West Virginia - the numbers just have - great Governor of West Virginia. The numbers just come out - West Virginia had the biggest percentage increase in GDP of any state in our union except for Texas; it was second. How's that? How's that? They said, oh coal will never come back. Now we're opening up lines and we're drilling and we're doing all sorts of things. And the windmills are wonderful, but you know what, when the wind doesn't blow they really do cause problems. We have no energy this week - well, hopefully the wind will start blowing pretty soon. We are pursuing American energy dominance. And by the end of this year we will be totally self-sufficient. We will be major exporters of energy. We will take in massive amounts of money. We will start paying off our \$20 trillion in debt. [Applause] My administration is also leading the charge to rebuild our military. Our military, right? [Applause] We're supporting our troops and their wonderful military families, including those right here in the cradle of Naval aviation, the home of the legendary Blue Angels, people don't realize that. [Applause] By the way, can those people fly or what? How about Naval Air Station Pensacola? Good place, right? And it's really time. This isn't a Republican thing or Democratic thing or anything. It's time for Democrats in Congress to drop the threats of shutdowns of government and support a clean appropriations bill that fully supports and funds our military. We can't play games anymore. [Applause] We can't play games. And don't forget, we make the greatest missiles, the greatest military equipment anywhere in the world. Nobody is close. And what does it also mean? And to me it's secondary in this case, but what does it also mean? Jobs, jobs. But we are building our military stronger and by the time I decide to go off into the wild blue yonder, you're going to have the strongest military this country has ever had by far. [Applause] It's wrong and dangerous for congressional Democrats to hold troop funding hostage for amnesty, letting people pour into our country, we don't know who they are. We don't know where they come from. You saw what happened with beautiful Kate Steinle. This guy, he said he didn't know it was a gun. Oh, he didn't know. Oh, he didn't know. He didn't know it was a gun. That's his new line. Do you notice the federal government came in and brought another suit against him, right? [Applause] We'll come after him. That was a total miscarriage of justice. He didn't know it was a gun. The real changes that we are facing are the drugs and the gangs pouring into our country, the millions of people overstaying their visas, chain migration that cost taxpayers billions and billions of dollars, and sanctuary cities that set free violent criminal aliens all over our country and protect them. [Crowd Booming] People come in and they are not necessarily good like the man that ran over -- the animal that ran over many people in New York City the other day. You saw that, two months ago. He runs over people. Goes on a beautiful -- I know it so well, the West Side Highway. I know it so well. So beautiful. People running, jogging, trying to get in shape. He killed many people, ran them over. Chain migration. According to chain migration, he may have as many as 22 to 24 people that came in with him, his grandfather, his grandmother, his mother, his father, his brothers, his sisters. We have to end chain migration. We have to end chain migration. [Applause] I mean, you go to Kate, who was such a beautiful young American woman, killed on a pier in San Francisco in the prime of her life. She was killed by this guy, an illegal alien who had been deported five times and convicted seven times of felonies. I guess they weren't allowed to mention this in court. They weren't allowed to mention. Don't worry, we're right in there. But he was roaming free to threaten innocent Americans like Kate because of our porous borders and because San Francisco is a sanctuary city. In other words, a city run by politicians who would rather protect criminal aliens than American citizens. I mean, let me give you another example. The city of Chicago, what the hell is going on in Chicago? There are those that say that Afghanistan is safer than Chicago, OK? What is going on? Do you know what's wrong with Chicago? Weak, ineffective politicians, Democrats that don't want to force restrictions. And don't -- and by the way, Chicago, for those of you that are going to say, guns, guns, Chicago has the toughest gun laws in the United States, OK? Just in case you were thinking about -- you know, they immediately say, oh, you have got to take away -- well, Chicago has the toughest gun laws in the United States. So we're asking Democrats in Congress to cease their obstruction and do the right thing, end sanctuary cities. [Applause] Stop the carnage. Save the innocent lives. We have also begun the process of building the wall on the southern border. [Applause] And we are empowering our immigration officers to do their job. And they are doing a great job. ICE and Border Patrol agents are working so hard, and they endorsed us for the election. I say us, because it's really not me, it's us all together, it's us, it's us, we are a group, a big group, a big beautiful group. All you have to do is look around here. [Applause] Just look around [Applause] I just

said just look around, and the first thing I see is that big beautiful merry Christmas sign. I mean, can you believe it? [Applause] Remember I told you two years ago when we started, I said -- and it was summer, but I said, you know, we are going to say merry Christmas, but let me just tell you, the department stores, right, they had the beautiful red walls, they had snowflakes all over the place, they had everything. Only one thing missing, the words merry Christmas. They are using those words again. Do you notice? They are using those words again. [Applause] But one by one we are finding the illegal alien drug dealers, the gang members, the thieves, the criminals, and the killers preying on our children, preying on everybody, and we are throwing them the hell out of our country or we are putting them in prison. [Applause] And because our borders now are strong as they should be, and they are going to get much stronger, we will not let these people back in. [Applause] It's time for Congress to adopt a pro-American immigration agenda. Every member of Congress should be asked where they stand on these issues. Blocking funds to sanctuary cities. I don't want to give sanctuary cities money. Passing Kate's law to put repeat offenders behind bars for a long period of time. [Applause] Increasing the number of ICE officers, who are fantastic, and Border Patrol officers so we can dismantle vile criminal gangs like, as I said, MS-13, animals. [Applause] Ensuring that new immigrants to our country are financially self-sufficient and will not be on welfare the day they come in. [Applause] And as I said, ending chain migration. We want a system that is merit-based. They come in on merit. They don't come in a lottery system. How about the lottery system, folks? Do you see that? [Crowd Booming] That's the guy in New York City, the lottery system, where they put names in a bin. You know, you think these countries are legit when they do their lottery system. So what they do, I would say, but more than just saying, they take their worst and they put them in the bin. And then when they pick the lottery, they have the real worst in their hands, oh, here we go. And we end up getting them. No more lottery system. We are going to end that. We have already started the process. [Applause] We want people coming into our country who love our people, support our economy, and embrace our values. It's time to get our priorities straight. This guy is screaming, we want Roy Moore. He is right. [Applause] Democrats in Congress want open borders, higher taxes, and government-run health care that doesn't work. Costs a fortune, doesn't work. They are soft on crime. And they want to suffocate our economy with socialist-style regulation. Raise your taxes through the sky, they don't want to vote for our tax cuts because they want the -- they want tax increases. That's why we need a Republican in the House. We need a Republican in the Senate. We need more of them. [Applause] And by the way, just so I can satisfy this gentleman out there. How many people here are from the great state of Alabama? [Applause] Whoa. Well, I have to say this, look, I have to. We have to be fair. So did you see what happened today? You know the yearbook? Did you see that? There was a little mistake made. She started writing things in the yearbook. Oh, what are we going to do? Gloria Allred, anytime you see her you know something's going wrong. We cannot afford -- this country, the future of this country, cannot afford to lose a seat in the very, very close United States Senate. We can't afford it, folks. We can't. [Applause] We can't afford to have a liberal Democrat who is completely controlled by Nancy Pelosi and Chuck Schumer. We can't do it. Can't do it. [Crowd Booming] His name is Jones and he is their total puppet. And everybody knows it. He will never ever vote for us. We need somebody in that Senate seat who will vote for our make America great again agenda, which involves tough on crime, strong on borders, strong on immigration. We want great people coming into our country. Building the wall. Strengthening our military. Continuing our great fight for our veterans. I love our veterans. [Applause] We -- right? We love our veterans. We want conservative judges like Judge Gorsuch on the Supreme Court. Doing a great job, too. We want people that are going to protect your gun rights, great trade deals instead of the horrible deals. [Applause] And we want jobs, jobs, jobs. So get out and vote for Roy Moore. [Applause] Do it. Do it. Do it. Working with Republicans in Congress, we have already signed 88 pieces of legislation. We get no credit. They always say, well, you know -- President Trump really needs this tax bill, because he hasn't passed any legislation. Well, so far, in 10 months, we've passed more during this period of time than any other President in the history of our country. And the second -- let's call runner-up -- is Harry Truman was second. So we passed all of this, including, including the long-awaited Veterans Accountability Act. You know what that is, right? David Shulkin, our Secretary, has done a great job. The VA was a mess, so many parts of our country... Hey, they left me with North Korea. They left me with a mess in the Middle East. They left me with Afghanistan, which is -- was a disaster. We make a lot of progress in Afghanistan just like we have with getting ISIS out of Iraq, getting ISIS out of Syria, knocking the hell out of them. They left me with a mess. But we're cleaning up the mess, we're cleaning it up, and with time, we'll have it spinning like a top. For decades, veterans have asked for a law to hold corrupt and incompetent employees and the VA accountable, but it never happened. Then, we got it done, this piece of legislation... for decades, they wanted it. You couldn't fire anybody at the VA if they were sadists, if they were horrible people, if they wouldn't show up to work, you couldn't fire anybody. Now, we look at them, they're no good, we say you're fired, get out of here. [Applause] And I must tell you: I can't believe that Arnold Schwarzenegger bombed so badly on The Apprentice. My poor, beautiful, show. It was so successful. We get a big movie star and he can't pull it off. We are respecting Americans heroes, and rebuilding America's standing. All over the world, they're respecting us again. When I came into office, we inherited all of these problems, all around the globe. Years of America projecting weakness, not going over the red line. If we went over the red line, you wouldn't have had Russia and Iran going into Syria, but you know what? He said: you're never going to do it. Don't ever violate the red line. They violated the red line. He did nothing. So Russia went in, and Iran went in. But you know what? We had 59 missiles -- and they use gas -- and we said you can't do that and we hit them -- our great military -- from ships 700 miles away, 59 missiles and 59 perfect hits. [Applause] And we've stopped apologizing for America, and we have started defending the reputation of the United States, finally. [Applause] To prevent Iran from ever obtaining a nuclear weapon, I recently declined to certify the disaster known as the Iran nuclear deal, a total disaster. How about that deal? How about that thing? We gave them a \$150 billion right? We gave them a \$150 billion, but here's the thing... smaller money, but think of this. We gave them \$1.8 billion in cash. I mean, I like green. In fact, it was so much cash, that they had to go to other countries also because we didn't have enough cash in Washington DC. They loaded up planes with cash. Can you believe, the president has that kind of power. I don't want to ever try it. No, but can you believe that the presidency has the power to send bushel loads of cash to Iran, probably for hostages, probably -- \$1.8 billion, probably for hostages. So sad, We don't do that. We don't do that anymore. As part of the campaign of maximum pressure on the vile dictatorship of North Korea, we have imposed the toughest ever sanctions passed by the United Nations Security Council, and we have a lot of other sanctions. But you know, I don't know that sanctions are going to work with him. We got to give it a shot. You know, we'll see. Who knows? I just tell you folks, you're in good hands. That's all I can say. That's all I can say. [Applause] And I've strengthened our relationships with America's allies, and asked other NATO members to pay their fair share. And now the money is pouring in. The money is pouring in. Remember I went to NATO a year ago and these fake people back here, they were saying Donald Trump's performance at NATO was unacceptable. You know why? You know why they said it? Because I told the people of NATO standing right behind me, while they were standing behind me, they've been delinquent, they haven't been paying. I said you go to pay, you gotta pay, you gotta pay. And now they've taken in, because of that -- and I guess I applied if you don't pay we're out of there, right -- and I took more heat from the press. They said: Donald Trump was rude to our allies. Well, they're rude to us when they don't pay, right? [Applause] They're rude to us. So we'll have a nation that doesn't pay. Then the nation gets frisky with whoever, Russia. So we have a nation doesn't pay. The nation gets aggressive. We end up in World War III for somebody that doesn't even pay. You think they think we're geniuses? So here's what's happened. In the last short period of time, Stoltenberg, who's the head of NATO, he's my biggest fan in the whole world, because he couldn't get anybody to pay. We've taken in \$11 billion extra, and going into 2020 or 2021, we will have taken in \$33 billion extra, and they aren't even doing what they should be doing, just so you understand, it's terrible, the way our country has been disrespected. But we will be disrespected no longer, okay? [Applause] Now tomorrow morning, you'll see from these fakers back there, Donald Trump again disrespects NATO. I don't disrespect them. I think NATO's wonderful. But you know what, we're paying for. 80% of NATO, could be higher. They say 72%. So we're paying for 80% of NATO. Now I can only tell you one thing: it helps them a hell of a lot more than it helps us, okay? And they should pay. Germany's paying 1%. We're paying 4%. Explain that one to me, right? And Germany has unsustainable cash flow. I read a report. Their cash flow is unsustainable. So I said to Angela, I said, Angela send a little of that cash flow our way. And she said, but Donald -- because we're protecting them -- we have 40,000 soldiers in Germany. Nobody even knows that. I said, Angela, let us have some. She said Donald, the German people wouldn't be happy with that. I said, well, you know what the American people aren't happy with the way we have it now, okay? [Applause] The American people aren't happy. And at home, we're restoring the rule of law. We're interpreting the Constitution, as written, defending the Second Amendment. You'll keep your guns. And protecting religious liberty. We are protecting religious liberty, and we're getting rid of the Johnson amendment. We've stopped the government's attacks on our Judeo-Christian values, because we know that families and churches, not government officials, know best how to create a strong and loving

community. We know that. Getting rid of the Johnson amendment, study that up. That's a big thing. That's a big thing. A lot of people are very, very happy about that. We know that parents, not bureaucrats, know best how to raise their children, and to nurture their children. [Applause]. And, above all else, we know this, America, doesn't worship, government, we worship God. [Applause] All of us here tonight are united by the same values. We love our country, and we are proud of our history. We are so proud of this country. We honor our heritage and we treasure our freedom. We support our incredible men and women of law enforcement. [Applause] We believe the United States military is the greatest force for justice in the history of the world, and we are going to take care of it, and we are going to properly fund it, and we are going to have the finest weapons, because when we do all of that we are much, much safer, and far less likely to have to use them. It's amazing how that works, isn't it? We believe that every American should stand for the national anthem -- [Applause, Crowd chants USA] -- and we proudly pledge allegiance to one nation under God. [Applause] Our rights come from our Creator, and no earthly force can ever take those rights away, and they never will. That is why my administration is taking power back from global bureaucrats and returning that power back to the American people, and you see it all the time. You see it economically, you see it at the church. You see it in every different way. We don't sing a global anthem, our troops don't wear a foreign uniform and we will never surrender our rights to international tribunals. We won't do that. [Applause] We proudly sing the star-spangled banner. Our brave troops, fight and die for the red, white and blue, and we protect and preserve the American Constitution that we cherish. I've said it so often that my job is not to be President of the world. My job is to be President of the United States of America. [Applause] Yet there are powerful forces, in Washington, trying to sabotage our movement. These are bad people. These are very, very bad and evil people. They know who they are. These are the people who made their money, their names, their careers, their power off the corrupt and broken system, and they liked it the other way. So they will do anything, at any time, and they'll never stop. But you know what we're stopping them. You're seeing that right now, you're seeing that right now, we're stopping them. It's corrupt, it's rigged and we're stopping them. They will lie and leak and smear, because they don't want to accept the results of an election where we won by a landslide. Do you remember? There is no way for Donald Trump to get 270. There, we heard that for week after week -- what they're doing is the fake news, it's called suppression. Just like they give the fake polls, I call it fake news, fake polls. A poll came out today on CNN. Such a fake. A poll came out. They will have to apologize for that one. Another one came out that was through the roof but it wasn't on CNN. But I will tell you, look, look, we need honesty. We need some love in the country. I would love to bring both sides together if that's possible. There is a lot of hatred out there. But I would love to be able to bring both sides together. [Applause] But remember this, the same failed voices in Washington who opposed our movement right from the beginning are the same people who have undermined the credibility of our government institutions. They are the same ones who gave us one terrible trade deal after another, and one foreign policy blunder and disaster after another, with no accountability and no apology. The Washington insiders who oppose our movement are the same people who sacrificed our sovereignty, our wealth, our borders, look at our country, we owe \$20 trillion, think of it. We have spent, as of two months ago, almost \$7 trillion in the Middle East. And do you know what we have? We have nothing. It's worse than it was 17 years ago when they started. These are the people that fight us. And if I want to fix up the roads in this country, they say, oh, well, we don't want to spend money for that, it's a tiny fraction. We could have, with that 7 trillion, not billion, \$7 trillion, we could have rebuilt our country three times over. They've had their chance at running this country and they failed. So I think I speak for everyone here tonight, and I appreciate your being here, especially in numbers where you can't even get into the corners, when I proudly say... [Applause] You know the beautiful thing, I was telling some of my friends backstage, the amazing thing about this and all the people that couldn't get in, when an election is over, they tell me, what do I know? I wasn't -- I have been a politician for a little more than two years. So what do I know? Although I'm doing a good job, I guess, right? I'm president. I'm President. [Applause] But -- but, it's sort of incredible, you look at everybody, you see everybody, you see the love in the room, and you see what's going on with everything. And you say to yourself, this is truly a country that respects -- they say that when you don't have an election -- the election is over. There is no competition. There is no nothing. Although we always have the deep seats inside. But, it's over. They say you can't draw a crowd. In other words, when an election ends, you can't draw flies. They say -- look at this place. Look at this place. [Applause] I will never forget on the trail, I would go, we would have 20,000, 25,000. We went to Mobile, Alabama, we had 49,000 people. [Applause] We went to Michigan the night of the election. I got there, started speaking at 12:30 in the evening. That means it was already Election Day. We had 32,000 people there. Michigan hadn't been won in many, many years by a Republican. Hillary Clinton went there in an emergency because she was told that day that she was doing badly in Michigan. She went there. She had a crowd of like 600 people. I had 32,000 people at 1:00 in the morning. I said, why are we not going to win? Why are we not going to win? And we won. [Applause] We went to Wisconsin, hasn't been won in many, many years. Do you remember that? We went to Wisconsin, hadn't been won in many, many years. She decided not to go. Thank you very much, Hillary. [Laughter] I guess the Russians advised her not to go, you know. [Laughter] But she decided not to go. And her husband supposedly said, you better go to Wisconsin. You better go to Michigan, I don't like what I'm seeing, I was in Michigan and I was driving down the street and they have a Trump/Pence sign on every single lawn all the way to the arena, you had better go there. [Applause] And she said no, no, no, my bunch of dummies -- bunch of dummies. You know, I don't get credit for one thing. It always bothered me. I spent a tiny fraction of what she spent and we won. You know, the old days, if you spend less and you win, that's a good thing. Over here, you don't get any credit for it. We spent 25 or 30 percent. She was at, what number, 2.2 billion, I understand, 2.2 billion. Do you think -- do you think those consultants made a couple of bucks on that election? Didn't she spend 12.4 million on a dossier that was a total phony, right, 12.4 million? [Crowd Boos] Boy. So we spent a tiny fraction, think of it, a tiny fraction of the money and we won. But we won because of people like you. [Applause] Because of people like you. And I will tell you what, if the Democrats had won that election, instead of being up 39 points in your 401(k)s, you would be down 50 percent from Election Day, I'm telling you, because it was ready to go down the tubes. I will tell you that. [Crowd calls out] Thank you, man. So we don't need the advice from the Washington swamp. We need to drain the swamp and we're going to do it and we are doing it. [Applause] That's why I don't really care about all the bitter attacks, all the phony stuff going on. Big media, special interests, phonies all. They are not here to protect you. They are only here to preserve and to protect themselves. And you are finding that out. I had a great life before I did this. Oh, think of where I would be right now if I didn't do this, I'd be very happy, believe me. But I wanted to do it, and I wanted to give back, and I saw what was happening, and I've always been good at doing the money thing. I saw what was happening to our country, and it wasn't good. It wasn't good. [Applause] I took this job on behalf of the forgotten men and women of our country. But guess what? They are forgotten no more. [Applause] No more. People came out of areas -- you know, they didn't think you existed. You know that, right? Remember? And do you remember the word deplorable? How brilliant was that? I was watching her with that speech. And she was reading a teleprompter, yet. And she said "deplorables." And I said She is talking about a lot of people. Little did I know, I was right. That thing blew up. That was one of the reasons she lost. And now we are all proud deplorables. We're proud deplorables. [Applause] Very proud. Your voice will never ever be ignored again. They are all going to be coming after you for every election from now on in this country. You will never be ignored again. Your dreams are my dreams. Your hopes are my hopes. And your future is what I am fighting for each and every day. [Applause] We're just getting started. We're on the verge of passing that wonderful, beautiful tax cut. It's the biggest in the history of our country. It doubles the amount of income taxed at the rate of zero. It lowers tax rate. It expands the child -- you know that, the child tax credit so broadly. It provides relief from the estate tax, also known as the death tax. It cuts small business taxes. It reduces the corporate rate, very importantly, from 35 percent, which is the highest in the industrialized world, all the way down to 20 percent. You're going to have new companies coming in. You're going to have jobs, jobs, jobs. And it brings corporate money from overseas back where it belongs. And we're talking about possibly in excess of \$4 trillion that we can finally bring back. The typical family of four earning \$75,000 will see an income tax cut of up to \$2,000 cutting their tax bill in half. People don't know it. And they don't want to report it. These people don't want to report it. Our business tax cut is expected to raise annual income for the typical household by more than \$4,000. They don't want to tell you that. They don't want to tell you that, fake news. Our plan means more companies will move to America, stay in America, and hire in America. We want every American to know the dignity of work, the pride of a paycheck, and the satisfaction of a job well done. [Applause] The Democrats in Washington want to grow our welfare rolls that you're going to pay for. They want to grow all sorts of things that you don't want to even think about. I want to grow our employment rolls with great jobs, well-paying jobs, jobs of dignity, jobs where people love waking up in the morning and going to work. And that's happening. So we're going to lift our people from welfare to work, from dependence to independence, and from

poverty to prosperity. We will restore hope to our struggling rural communities and to our inner cities where we are making progress. We are making progress. We will build new roads and bridges and tunnels and highways all throughout our land. Factories will come roaring back to life. New works of iron and steel will be forged from fire and from spark and from our country. And they will be made with American muscle, American hands, and most importantly, American heart. [Applause] And these creations will be branded with a very simple but very beautiful phrase, made in the USA. [Applause] We have it in our power to build this future together, a future of patriotism, prosperity and pride. We all share one home and one glorious destiny. And whether we are young or old -- I feel very young, do you feel young? [Applause] Whether we are from the city or the country, and whether we are black or brown or white, you've heard it before, we all bleed the same red blood. [Applause] We all salute the same great American flag. [Applause] And we are all made by the same almighty God. [Applause] We are the nation that dug out the Panama Canal, won two world wars, put a man on the moon, and brought communism to its knees. [Applause] As long as we have the courage of our convictions, and the strength to see them through, then there is no goal beyond our reach. As long as we are true to our values, loyal to our citizens, and faithful to our God, then we will not fail. [Applause] This is your land. This is your home. And it's your voice that matters the most. So speak up, be heard, and fight, fight, fight for the change you have been waiting for your entire life, for the change that you already see happening. Our revolution didn't end on November 8th. That was just the beginning. [Applause] The greatest adventure still lies ahead. Never give in, never give up, never back down, and never ever stop dreaming. [Applause] Because we are Americans and the future belongs to us. The future belongs to all of you. So with American pride swelling in our hearts and American courage stirring in our souls, I say these words tonight, together we will, indeed, make America strong again. [Applause] We will make America proud again. [Applause] We will make America wealthy again. [Applause] We will make America safe again. [Applause] And put it all together and what do we have? We will make America great again. Thank you. God bless you. God bless America. Thank you, everybody.

3. Named Entity Recognition — Who Gets Mentioned

Runs spaCy's `en_core_web_sm` NER model across every transcript and pulls out all `PERSON` entities, then ranks the 20 most frequently mentioned names (after filtering out non-person artifacts like "Audience Boos" that the NER model occasionally mis-tags).

```
In [ ]: import pandas as pd
import spacy
from collections import Counter

# Load SpaCy's English model
nlp = spacy.load("en_core_web_sm")

# Assuming 'df' is your dataframe and 'Transcript' contains the text
texts = df['Transcript'].dropna().tolist()

# Extract only PERSON entities
person_entities = []
for text in texts:
    doc = nlp(text)
    person_entities.extend([ent.text for ent in doc.ents if ent.label_ == "PERSON"])

# Count the frequency of PERSON entities
person_entity_counts = Counter(person_entities)

# Remove unwanted entities (e.g., "Audience Boos")
exclude_entities = {"Audience Boos", "Audience Boos"}
filtered_person_entity_counts = {k: v for k, v in person_entity_counts.items() if k not in exclude_entities}

# Convert to a dataframe for better viewing
person_entity_df = pd.DataFrame(filtered_person_entity_counts.items(), columns=["Entity", "Frequency"])
person_entity_df = person_entity_df.sort_values(by="Frequency", ascending=False)

# Display the top 20 PERSON entities
top_20_person_entities = person_entity_df.head(20)
print(top_20_person_entities)
```

	Entity	Frequency
3	Biden	340
54	Trump	309
1	Joe	283
14	Donald Trump	253
49	Obama	195
2	Joe Biden	193
26	Bernie	163
37	Mike	154
68	Nancy Pelosi	139
9	Hillary	123
125	God	114
324	David	95
107	Jim	92
161	Hillary Clinton	81
29	Ron	79
28	Hunter	78
35	John	76
544	Brett Kavanaugh	72
246	Chuck Schumer	70
33	Matt	68

4. Tracking Democratic-Figure Mentions Over Time

Counts, per transcript, how many `PERSON` entities match a hand-curated list of Democratic figures (Biden, Obama, Pelosi, etc.), then plots that count against rally date.

Note: this cell references a `dates` variable that isn't defined anywhere earlier in the notebook — it likely needs `dates = df['Date'].tolist()` (or similar) added before `mentions_df` is built, or the cell will raise a `NameError` on a fresh run.

```
In [ ]: # List of Democratic party names (can be expanded)
democratic_names = {
    "Biden", "Joe", "Joe Biden", "Obama", "Bernie", "Hillary",
    "Nancy Pelosi", "Hillary Clinton", "Hunter", "Chuck Schumer"
}

# Count Democratic mentions per transcript
democratic_counts = []
for text in texts:
    # Tokenize each transcript and count Democratic mentions
    doc = nlp(text)
    entities = [ent.text for ent in doc.ents if ent.label_ == "PERSON"]
    democratic_count = sum(1 for entity in entities if entity in democratic_names)
    democratic_counts.append(democratic_count)

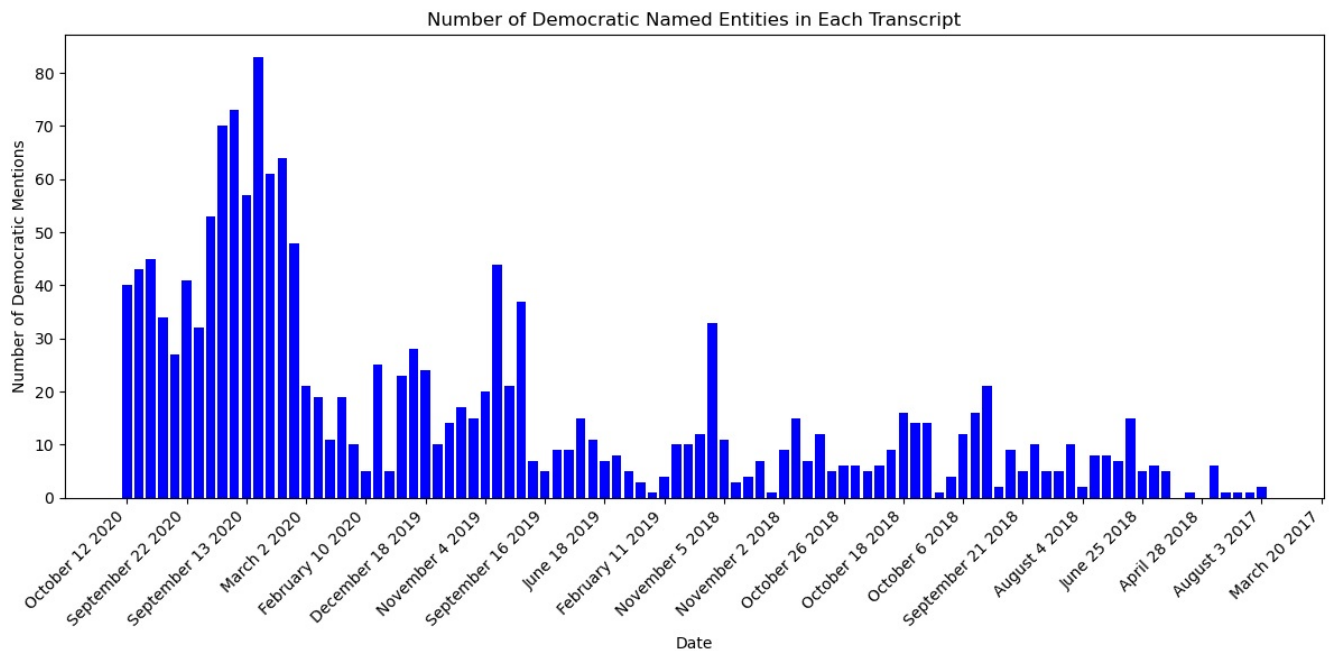
# Step 5: Create a DataFrame for Democratic mentions
mentions_df = pd.DataFrame({"Date": dates, "Democratic Mentions": democratic_counts})

import matplotlib.pyplot as plt

# Step 6: Plot the number of Democratic mentions with improved x-axis formatting
plt.figure(figsize=(12, 6))
plt.bar(mentions_df['Date'], mentions_df['Democratic Mentions'], color='blue')
plt.xlabel("Date")
plt.ylabel("Number of Democratic Mentions")
plt.title("Number of Democratic Named Entities in Each Transcript")

# Adjust the x-axis ticks to show only every nth date
n = 5 # Show every 5th date (adjust as needed)
plt.xticks(ticks=range(0, len(mentions_df['Date']), n), labels=mentions_df['Date'][::n], rotation=45, ha='right')

plt.tight_layout()
plt.show()
```



5. Transcript Similarity Analysis (TF-IDF + Cosine Similarity)

Vectorizes every transcript with TF-IDF, computes pairwise cosine similarity between all 103 speeches, and visualizes the resulting similarity matrix as a heatmap — useful for spotting which rallies reuse very similar material.

```
In [ ]: import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
import matplotlib.pyplot as plt
import seaborn as sns

# Assuming 'df' is your dataframe and 'Transcript' contains the text
texts = df['Transcript'].dropna().tolist()

# Step 1: Vectorize the transcripts using TF-IDF
vectorizer = TfidfVectorizer(stop_words='english')
```

```

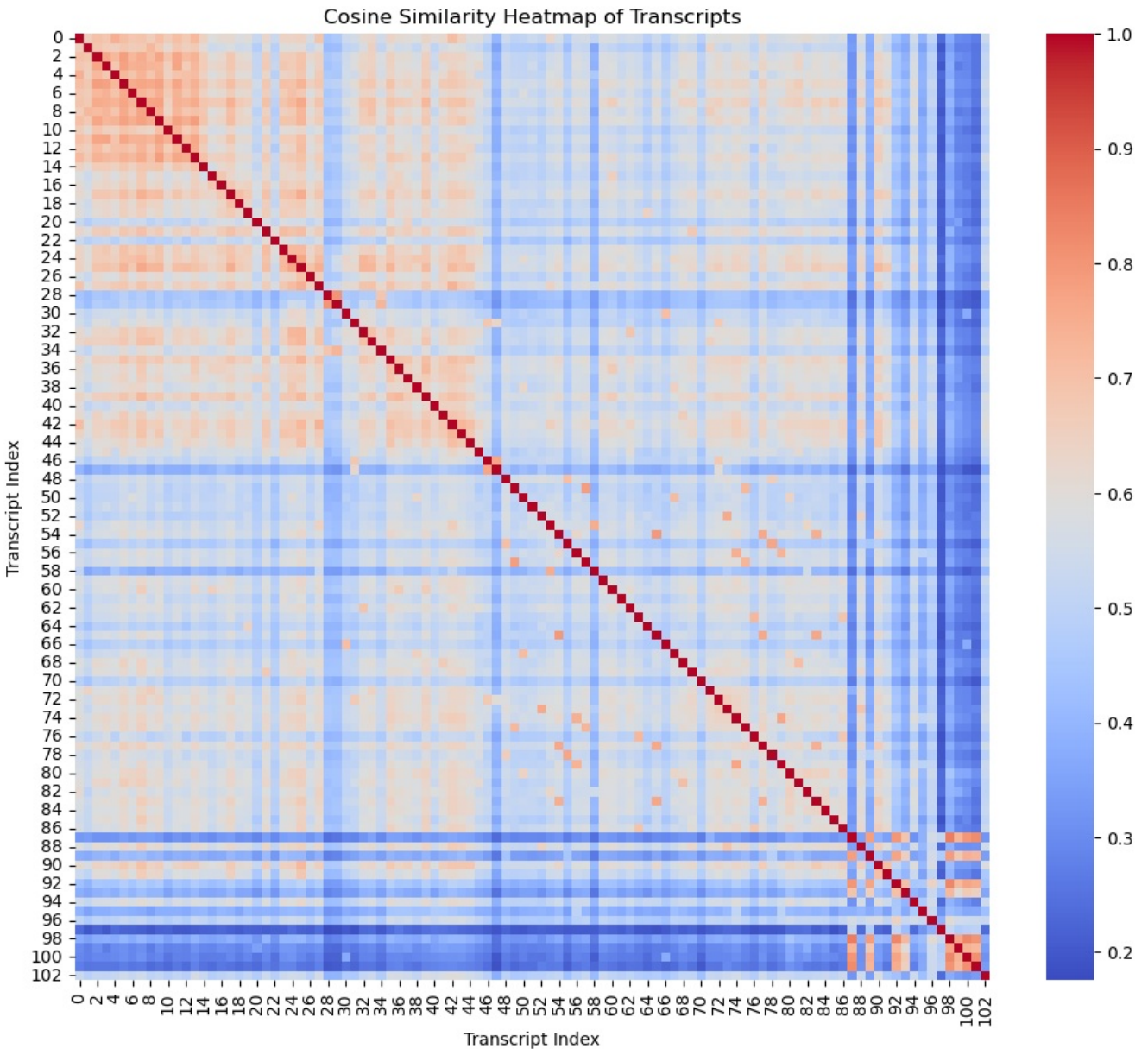
tfidf_matrix = vectorizer.fit_transform(texts)

# Step 2: Compute pairwise cosine similarity
cosine_sim_matrix = cosine_similarity(tfidf_matrix)

# Step 3: Create a DataFrame for better visualization
cosine_sim_df = pd.DataFrame(cosine_sim_matrix, index=df.index, columns=df.index)

# Step 4: Plot the heatmap
plt.figure(figsize=(12, 10))
sns.heatmap(cosine_sim_df, cmap="coolwarm", annot=False)
plt.title("Cosine Similarity Heatmap of Transcripts")
plt.xlabel("Transcript Index")
plt.ylabel("Transcript Index")
plt.show()

```



6. Extracting Audience / Crowd Reactions

Uses a regex to pull out every bracketed audience cue in the transcripts (e.g. [Audience Boos], [Audience chants "USA"]) along with the sentence that immediately preceded it, producing a Preceding Phrase / Crowd Reaction table.

Note: the regex catches *any* bracketed tag, including non-crowd markers like [Inaudible], so `reaction_df` will need a light filter if the goal is audience reactions specifically.

```

In [ ]: import pandas as pd
import re

```

```

# Assuming 'df' is your DataFrame and 'Transcript' is the column with the text data
def extract_reactions(transcript):
    # Find all instances of crowd reactions and the preceding text
    matches = re.finditer(r"(.*?)s*([^\]]+)", transcript)
    return [(match.group(1).strip(), match.group(2).strip()) for match in matches]

# Apply the extraction to each transcript
reaction_data = []
for transcript in df['Transcript']:
    reactions = extract_reactions(transcript)
    reaction_data.extend(reactions)

# Create a new DataFrame
reaction_df = pd.DataFrame(reaction_data, columns=['Preceding Phrase', 'Crowd Reaction'])

# Display the first few rows to verify
print(reaction_df.head())

```

```

          Preceding Phrase \
0 Hello, everybody. Hello, Orlando. Hello, Sanfo...
1 That's a lot of people. You know our competitio...
2 Thank you. Thank you. Thank you very much. Tha...
3 Thank you, man. Thank you. This is the most im...
4 He's got nothing going. If he wins, the radica...

          Crowd Reaction
0 [Audience chants "USA"]
1 [Audience chants "We love you"]
2 [Audience chants "USA"]
3 [Audience Boos]
4 [Inaudible]

```

```
In [ ]: reaction_df.shape
```

```
(2723, 2)
```

7. Sentiment Scoring of Crowd Reactions (SentiWordNet)

Applies SentiWordNet — a lexicon-based sentiment scorer — to the extracted Crowd Reaction text itself (e.g. scoring the words in "Audience Boos"), labels each reaction positive (1) or negative (0), and drops neutral/unscored ones.

```

In [ ]: import pandas as pd
        from nltk.corpus import sentiwordnet as swn
        from nltk.tokenize import word_tokenize
        from nltk.corpus import wordnet
        import nltk

        # Download necessary resources
        nltk.download('sentiwordnet')
        nltk.download('wordnet')
        nltk.download('punkt')

        # Function to get the WordNet POS tag
        def get_wordnet_pos(tag):
            if tag.startswith('J'):
                return wordnet.ADJ
            elif tag.startswith('V'):
                return wordnet.VERB
            elif tag.startswith('N'):
                return wordnet.NOUN
            elif tag.startswith('R'):
                return wordnet.ADV
            else:
                return None

        # Function to calculate sentiment score using SentiWordNet
        def analyze_sentiment_swn(crowd_reaction):
            tokens = word_tokenize(crowd_reaction)
            sentiment_score = 0
            count = 0

            for token in tokens:
                synsets = list(swn.senti_synsets(token))
                if synsets:
                    # Use the first synset's sentiment scores
                    synset = synsets[0]
                    sentiment_score += synset.pos_score() - synset.neg_score()
                    count += 1

            # Average sentiment score
            if count == 0:
                return None
            avg_score = sentiment_score / count

            # Assign sentiment label: 1 for positive, 0 for negative, drop neutral
            if avg_score > 0:

```

```

        return 1
    elif avg_score < 0:
        return 0
    else:
        return None

# Apply sentiment analysis to the 'Crowd Reaction' column
sentiment_reaction_df = reaction_df.copy() # Copy the original DataFrame
sentiment_reaction_df['Sentiment Label'] = sentiment_reaction_df['Crowd Reaction'].apply(analyze_sentiment_swn)

# Drop rows with neutral reactions
sentiment_reaction_df = sentiment_reaction_df.dropna(subset=['Sentiment Label'])

# Convert to integer for consistency
sentiment_reaction_df['Sentiment Label'] = sentiment_reaction_df['Sentiment Label'].astype(int)

# Display the updated DataFrame
print(sentiment_reaction_df.head())

```

```

[nltk_data] Downloading package sentiwordnet to
[nltk_data] C:\Users\Gilles\AppData\Roaming\nltk_data...
[nltk_data] Package sentiwordnet is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data] C:\Users\Gilles\AppData\Roaming\nltk_data...
[nltk_data] Package wordnet is already up-to-date!
[nltk_data] Downloading package punkt to
[nltk_data] C:\Users\Gilles\AppData\Roaming\nltk_data...
[nltk_data] Package punkt is already up-to-date!

```

```

        Preceding Phrase \
1 That's a lot of people. You know our competitio...
2 Thank you, man. Thank you. This is the most im...
3 He's got nothing going. If he wins, the radica...
4 . And if I don't always play by the rules of W...
5 How about where he gets caught again? He said,...

```

	Crowd Reaction	Sentiment Label
1	[Audience chants "We love you"]	1
3	[Audience Boos]	0
4	[Inaudible]	0
5	[Audience Boos]	0
6	[Audience Boos]	0

```
In [ ]: sentiment_reaction_df.shape
```

```
(1790, 3)
```

8. Feature Importance for Sentiment — Mutual Information

Loads a separate, pre-labeled dataset (`filtered_phrases_nouns_verbs.xlsx`, with `Filtered_phrase` + `label` columns) and uses TF-IDF + mutual information to find the words most associated with positive vs. negative sentiment.

Notes:

- This reads from a **hardcoded local Windows path** (`C:\Users\Gilles\...`), so it won't run outside that original machine — it'll need to point at wherever this labeled file lives.
- `df1` is a **different dataset** than `df/reaction_df` above; the notebook doesn't show how it was produced, so the link between the crowd-reaction sentiment work in Section 7 and this phrase-level labeled set isn't documented here.

```

In [ ]: import pandas as pd
import warnings

# Suppress all warnings
warnings.filterwarnings("ignore")

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.feature_selection import mutual_info_classif
import matplotlib.pyplot as plt

# Load the dataset
df1 = pd.read_excel(r'C:\Users\Gilles\Desktop\montr here I come\Previous ML and text mining projects winter 2024\NLP
projects\filtered_phrases_nouns_verbs.xlsx')

# Fill NaN values in the 'Filtered_phrase' column with an empty string
df1['Filtered_phrase'].fillna('', inplace=True)

# Vectorize the 'Filtered_phrase' column
vectorizer = TfidfVectorizer(max_features=1000) # Adjust max_features to suit your needs
X = vectorizer.fit_transform(df1['Filtered_phrase'])

# Get the sentiment labels
y = df1['label']

# Calculate mutual information for the features
mi_scores = mutual_info_classif(X, y)

```

```
# Get feature names
feature_names = vectorizer.get_feature_names_out()

# Pair feature names with scores and filter by class
positive_features = [(name, score) for name, score in zip(feature_names, mi_scores) if
df1[df1['Filtered_phrase'].str.contains(name)]['label'].mean() > 0.5]
negative_features = [(name, score) for name, score in zip(feature_names, mi_scores) if
df1[df1['Filtered_phrase'].str.contains(name)]['label'].mean() < 0.5]

# Sort the features by score for each sentiment
top_positive_features = sorted(positive_features, key=lambda x: x[1], reverse=True)[:20]
top_negative_features = sorted(negative_features, key=lambda x: x[1], reverse=True)[:20]

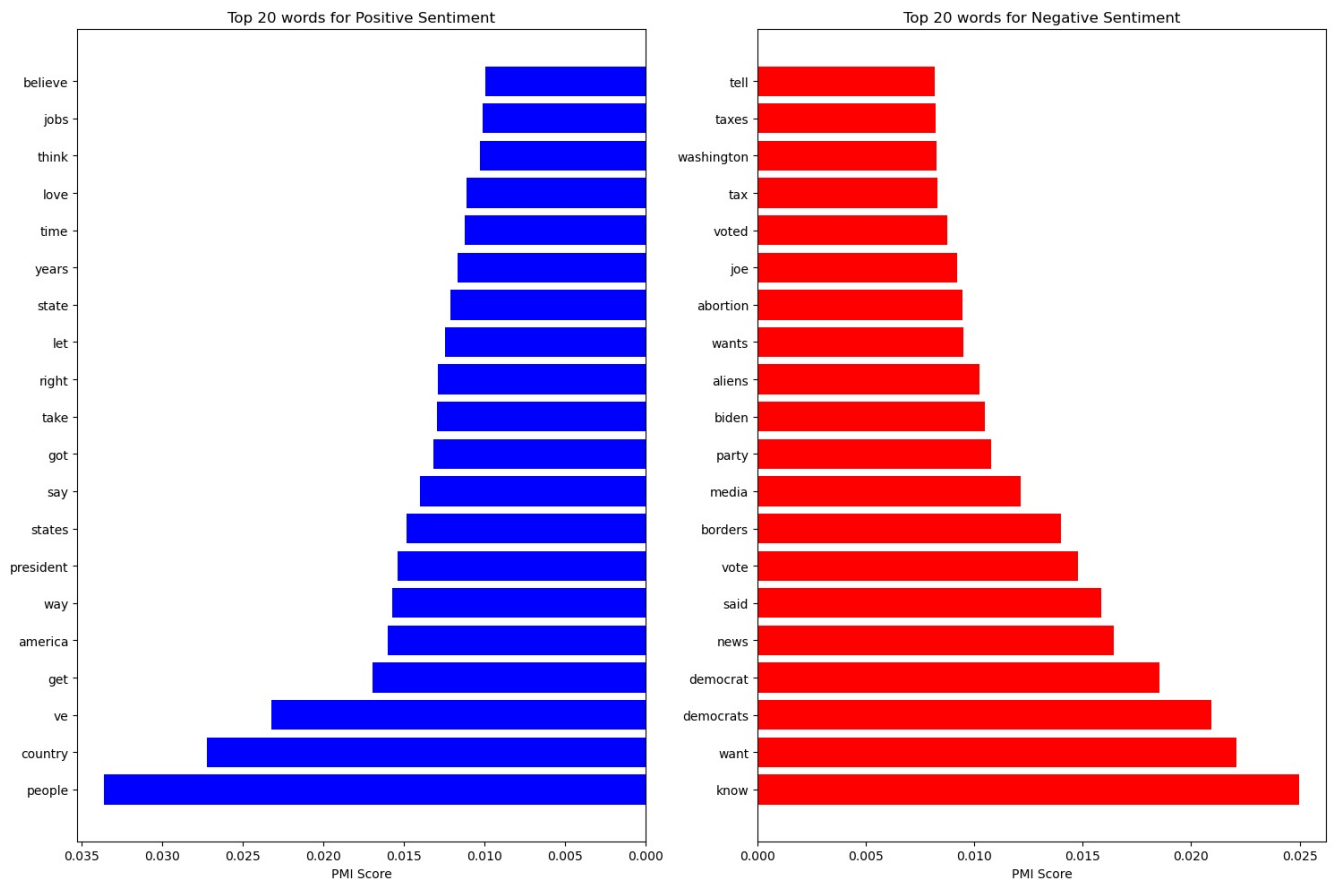
# Extracting the features and their scores
positive_words, positive_scores = zip(*top_positive_features)
negative_words, negative_scores = zip(*top_negative_features)

# Create subplots
fig, axes = plt.subplots(nrows=1, ncols=2, figsize=(15, 10))

# Plotting the positive sentiment words
axes[0].barh(positive_words, positive_scores, color='blue')
axes[0].set_title('Top 20 words for Positive Sentiment')
axes[0].invert_yaxis() # labels read top-to-bottom
axes[0].set_xlabel('PMI Score')

# Plotting the negative sentiment words
axes[1].barh(negative_words, negative_scores, color='red')
axes[1].set_title('Top 20 words for Negative Sentiment')
axes[1].set_xlabel('PMI Score')

plt.tight_layout()
plt.show()
```



9. Feature Importance for Sentiment — Chi-Square Test

Repeats the same idea as Section 8 with a domain-specific stopwords list removed first, then ranks words by chi-square score instead of mutual information.

Note: this cell calls `chi2(...)` but never imports it — it needs `from sklearn.feature_selection import chi2` added at the top of the cell.

```
In [ ]: # Define the stopwords that you want to remove
```

```

stopwords = {'know', 'want', 'wants', 'supports', 'crooked', 'party', 'straight', 'saw', 'moment',
             'left', 'guys', 'happened', 'idea', 'leader', 'lou', 'lies', 'gets', 'democrat',
             'trying', 'made', 'keep', 'need', 'telling', 'call', 'shut'}

# Function to remove stopwords from a phrase
def remove_stopwords(phrase):
    return ' '.join(word for word in phrase.split() if word not in stopwords)

# Fill NaN values in the 'Filtered_phrase' column with an empty string
df1['Filtered_phrase'].fillna('', inplace=True)
df1['Filtered_phrase'] = df1['Filtered_phrase'].apply(remove_stopwords)

# Vectorize the 'Filtered_phrase' column
vectorizer = TfidfVectorizer(max_features=1000) # Adjust max_features to suit your needs
X = vectorizer.fit_transform(df1['Filtered_phrase'])

# Get the sentiment labels
y = df1['label']

# Apply Chi-Square test
chi2_scores, p_values = chi2(X, y)

# Get feature names
feature_names = vectorizer.get_feature_names_out()

# Function to get the mean label for a given word
def mean_label(word):
    mask = df1['Filtered_phrase'].str.contains(word, na=False)
    return df1.loc[mask, 'label'].mean()

# Pair feature names with chi2 scores and filter by class
positive_features = [(name, score) for name, score in zip(feature_names, chi2_scores) if mean_label(name) > 0.5]
negative_features = [(name, score) for name, score in zip(feature_names, chi2_scores) if mean_label(name) < 0.5]

# Sort the features by chi2 score for each sentiment
top_positive_features = sorted(positive_features, key=lambda x: x[1], reverse=True)[:35]
top_negative_features = sorted(negative_features, key=lambda x: x[1], reverse=True)[:35]

# Extracting the features and their scores
positive_words, positive_scores = zip(*top_positive_features)
negative_words, negative_scores = zip(*top_negative_features)

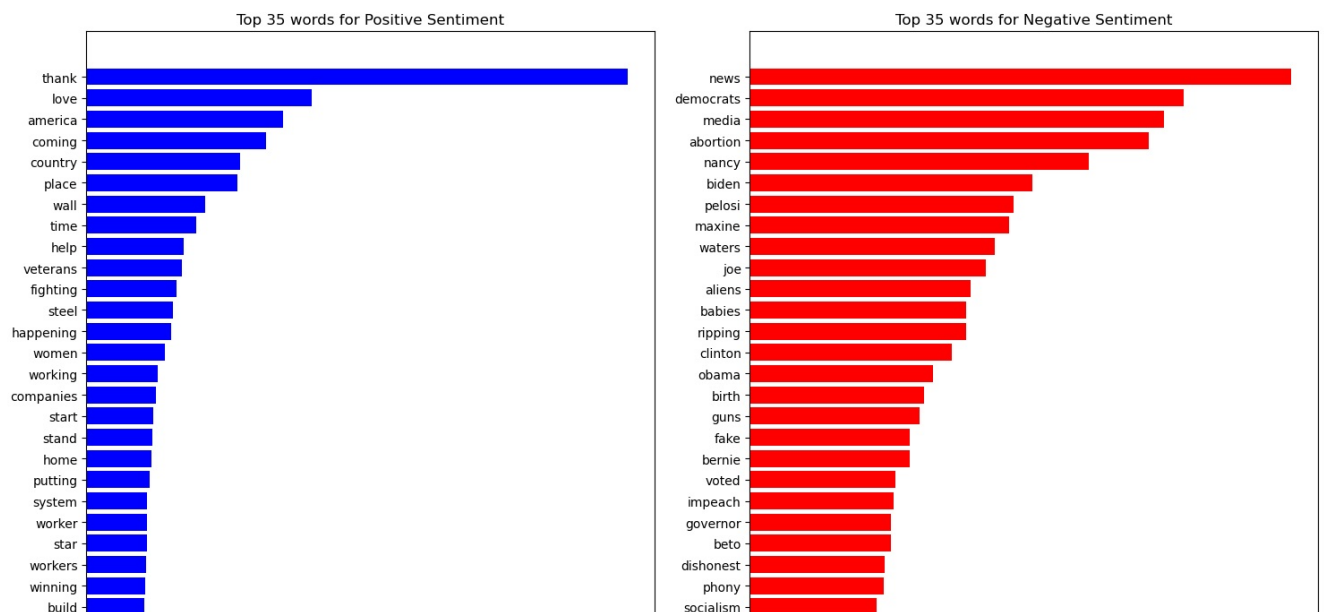
# Create subplots
fig, axes = plt.subplots(nrows=1, ncols=2, figsize=(15, 10))

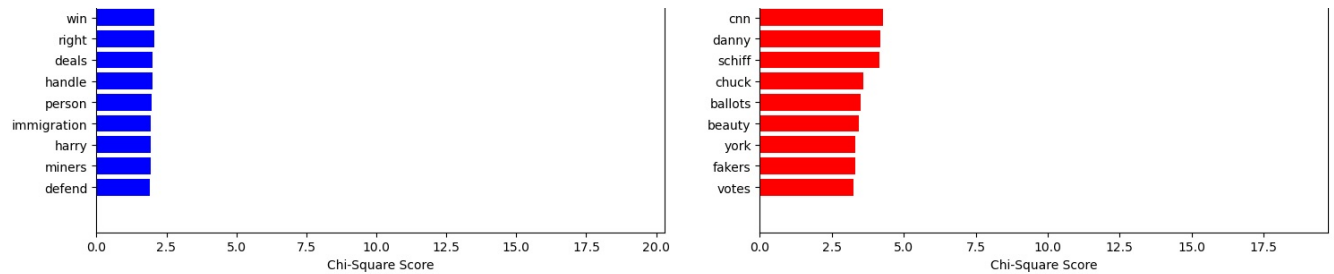
# Plotting the positive sentiment words
axes[0].barh(positive_words, positive_scores, color='blue')
axes[0].set_title('Top 35 words for Positive Sentiment')
axes[0].invert_yaxis() # Labels read top-to-bottom
axes[0].set_xlabel('Chi-Square Score')

# Plotting the negative sentiment words
axes[1].barh(negative_words, negative_scores, color='red')
axes[1].set_title('Top 35 words for Negative Sentiment')
axes[1].invert_yaxis() # Labels read top-to-bottom
axes[1].set_xlabel('Chi-Square Score')

plt.tight_layout()
plt.show()

```





10. Extracting Biden-Related Sentences

Splits every transcript into sentences and keeps only the ones mentioning "Joe", "Biden", or "Joe Biden", pooling them into `all_biden_sentences` — this becomes the input corpus for the topic modeling sections below.

```
In [ ]: import pandas as pd
import re

# Define a function to extract sentences containing keywords
def extract_sentences_about_biden(transcript):
    # Split the transcript into sentences using a regex
    sentences = re.split(r'(?<=[!?!]) +', transcript)
    # Filter sentences containing "joe", "biden", or "joe biden" (case-insensitive)
    biden_sentences = [sentence for sentence in sentences if re.search(r'\b(joe|biden|joe biden)\b', sentence, re.IGNORECASE)]
    return biden_sentences

# Apply the function to each transcript and aggregate all sentences
all_biden_sentences = []
for transcript in df['Transcript']:
    if pd.notna(transcript): # Check if transcript is not NaN
        all_biden_sentences.extend(extract_sentences_about_biden(transcript))

# Aggregate all sentences into a single string (optional)
aggregated_biden_sentences = " ".join(all_biden_sentences)

# Display the first few sentences as a preview
print("\n".join(all_biden_sentences[:10])) # Preview the first 10 sentences
```

You know our competitor, sleepy Joe, he had a rally today and practically nobody showed up. Sleepy Joe, but it's great to be back in my home state, Florida to make my official return to the campaign trail. Sleepy Joe Biden, not a nice guy, by the way, I have to be honest. No, Biden had a bad day. Now Biden had a bad day. Joe Biden is also owned by the radical globalists, the wealthy donors, the big money special interests who shipped away your jobs, shut down your factories, threw open your borders, and ravaged your cities while sacrificing American blood and treasure in this ridiculous -- endless wars, set of endless wars that we've been in for a long time, 19 years. Biden has made a corrupt bargain exchange for his party's nomination, he's handed control to the Socialists, the Marxists, and the left-wing extremists and you know that and he's got no strength left, he's got no power left. Sleepy Joe wants to quadruple your taxes, you know, wants to quadruple. And by the way, Obama and Biden knew everything that was happening, OK? Joe Biden and the Democrat Socialist, but that gives you another reason to go out and vote.

```
In [ ]: print(len(all_biden_sentences)) # Prints the number of sentences in the list
```

880

11. Topic Modeling — BERTopic

Cleans the Biden-related sentences (removing custom stopwords and keeping only nouns/verbs/adjectives/adverbs via spaCy POS tags), embeds them with a pretrained `all-mpnet-base-v2` sentence-transformer, and clusters them into topics with BERTopic — printing the resulting topic table and the top terms for the first 5 topics.

```
In [ ]: from bertopic import BERTopic
from sentence_transformers import SentenceTransformer
import spacy

# Assuming `all_biden_sentences` contains the extracted sentences
biden_sentences = all_biden_sentences # List of sentences mentioning "Joe Biden"

# Step 1: Define custom stop words
custom_stopwords = {
    "said", "think", "got", "let", "like", "did", "going", "remember", "know",
    "compared", "months", "day", "thing", "people", "bad", "right", "can", "joe",
    "biden", "had", "now", "say", "go", "gon", "just", "have", "here", "so", "do", "get", "as",
    "need", "done", "then"}

# Load SpaCy's English model
nlp = spacy.load("en_core_web_sm")

# Step 2: Preprocess sentences by removing stop words and grammar words
```

```

def preprocess_sentences(sentences):
    processed_sentences = []
    for sentence in sentences:
        doc = nlp(sentence)
        filtered_words = [
            token.text for token in doc
            if token.is_alpha # Only keep alphabetic tokens
            and token.text.lower() not in custom_stopwords # Exclude custom stop words
            and token.pos_ not in {"AUX", "CCONJ", "DET", "PRON"} # Exclude grammar words
            and token.pos_ in {"NOUN", "VERB", "ADJ", "ADV"} # Keep meaningful POS
        ]
        processed_sentences.append(" ".join(filtered_words))
    return processed_sentences

# Apply preprocessing
cleaned_biden_sentences = preprocess_sentences(biden_sentences)

# Step 3: Load the pretrained transformer model
embedding_model = SentenceTransformer('all-mpnet-base-v2')

# Step 4: Initialize BERTopic with the pretrained embedding model
topic_model = BERTopic(embedding_model=embedding_model, language="english")

# Step 5: Fit the BERTopic model
topics, probs = topic_model.fit_transform(cleaned_biden_sentences)

# Step 6: Get the topic information
topic_info = topic_model.get_topic_info()
print("Topic Information:\n", topic_info)

# Step 7: Get detailed information about the top N topics
for i in range(5): # Adjust range to see more topics
    print(f"\nTopic {i}:\n", topic_model.get_topic(i))

```

Topic Information:

	Topic	Count	Name \
0	-1	142	-1_destroy_opposed_charter_school
1	0	138	0_son_take_tell_good
2	1	116	1_left_police_radical_rioters
3	2	101	2_tax_ban_wants_vaccine
4	3	82	3_sleepy_endorsed_guy_better
5	4	78	4_jobs_supported_coal_american
6	5	48	5_____
7	6	38	6_years_last_damage_reversing
8	7	31	7_vote_voted_war_votes
9	8	23	8_call_percent_fairness_ran
10	9	19	9_country_lead_chairman_party
11	10	19	10_wins_simple_mob_very
12	11	17	11_slow_circles_little_smoother
13	12	15	12_agenda_made_list_seniors
14	13	13	13_debate_night_nomination_single

	Representation \
0	[destroy, opposed, charter, school, put, condi...
1	[son, take, tell, good, great, paid, doing, ev...
2	[left, police, radical, rioters, far, law, sup...
3	[tax, ban, wants, vaccine, allow, voted, impos...
4	[sleepy, endorsed, guy, better, slow, shouts, ...
5	[jobs, supported, coal, american, state, globa...
6	[, , , , , , , , ,]
7	[years, last, damage, reversing, spent, inflic...
8	[vote, voted, war, votes, forgiveness, begging...
9	[call, percent, fairness, ran, calls, times, m...
10	[country, lead, chairman, party, save, believe...
11	[wins, simple, mob, very, desperately, problem...
12	[slow, circles, little, smoother, wait, slower...
13	[agenda, made, list, seniors, manifesto, stori...
14	[debate, night, nomination, single, tell, time...

	Representative Docs
0	[plan destroy destroy protections pre existing...
1	[sleepy son getting money become take deal, ri...
2	[last week proudly accepted endorsement pro cr...
3	[decide quickly return record prosperity allow...
4	[sleepy, sleepy, sleepy]
5	[pledged end production American oil shale cle...
6	[, , ,]
7	[spent last years reversing damage inflicted l...
8	[vote, vote, vote]
9	[call, call percent, call percent]
10	[too weak lead country, lead country really be...
11	[wins wins, wins wins wins, wins wins]
12	[guys ever see sleepy little circles, brought ...
13	[agenda made, agenda made, agenda made]
14	[first lady actually came time watches goes ni...

Topic 0:

```
[('son', 0.07078190150614841), ('take', 0.055294534525272075), ('tell', 0.04469785603982017), ('good', 0.04305139695289015),
```

```
('great', 0.04195632369377114), ('paid', 0.0402694527117318), ('doing', 0.03785066601019808), ('even', 0.03729930823373019), ('see', 0.03618120508637101), ('energy', 0.03609620406664099)]
```

Topic 1:

```
[('left', 0.08317976108107136), ('police', 0.05196785673921021), ('radical', 0.05003446178989798), ('rioters', 0.04663018487939597), ('far', 0.041874667554988976), ('law', 0.040412826895476504), ('supporters', 0.03568327315555868), ('liberal', 0.03491172471044412), ('enforcement', 0.0331990797706246), ('wing', 0.032351281322338346)]
```

Topic 2:

```
[('tax', 0.05960332051336176), ('ban', 0.04835487977361925), ('wants', 0.043620782307690224), ('vaccine', 0.04085748180997102), ('allow', 0.0391539403615424), ('voted', 0.036977261003355896), ('impose', 0.03571463480789314), ('destroy', 0.03544188562499831), ('plan', 0.03464952369535339), ('economy', 0.034047901508309185)]
```

Topic 3:

```
[('sleepy', 0.3379032355360281), ('endorsed', 0.0506382339488843), ('guy', 0.050210687031589936), ('better', 0.04251038786563388), ('slow', 0.041403313166515646), ('shouts', 0.037978675461663225), ('stage', 0.037978675461663225), ('crazy', 0.03500869102972497), ('beat', 0.03500869102972497), ('see', 0.03249808840093204)]
```

Topic 4:

```
[('jobs', 0.0865264698234511), ('supported', 0.05286186632758752), ('coal', 0.04491203854995992), ('american', 0.041607639640460596), ('state', 0.04044098929177087), ('globalist', 0.03620460973006268), ('countries', 0.03485216931126122), ('foreign', 0.03368402891246994), ('pledged', 0.03265663370441705), ('manufacturing', 0.03150739593689895)]
```

12. Topic Modeling — NMF (Non-negative Matrix Factorization)

A second, lighter-weight topic modeling pass over the same cleaned Biden sentences: TF-IDF vectorization followed by NMF with a fixed `n_topics = 5`, printing the top 10 words per topic — useful as a classic-ML point of comparison against the BERTopic results above.

```
In [ ]: from sklearn.feature_extraction.text import TfidfVectorizer
        from sklearn.decomposition import NMF
        import pandas as pd

        # Assuming `processed_biden_sentences` contains the preprocessed sentences
        # Example: processed_biden_sentences = ["biden economy plan", "joe vaccine rollout", ...]

        # Step 1: Convert sentences into a TF-IDF matrix
        tfidf_vectorizer = TfidfVectorizer(max_features=1000, stop_words='english') # Adjust max_features as needed
        tfidf_matrix = tfidf_vectorizer.fit_transform(cleaned_biden_sentences)

        # Step 2: Apply NMF for topic modeling
        n_topics = 5 # Number of topics
        nmf_model = NMF(n_components=n_topics, random_state=42)
        W = nmf_model.fit_transform(tfidf_matrix) # Document-topic matrix
        H = nmf_model.components_ # Topic-word matrix

        # Step 3: Extract the top words for each topic
        feature_names = tfidf_vectorizer.get_feature_names_out()

        def display_topics(H, feature_names, n_top_words):
            for topic_idx, topic in enumerate(H):
                print(f"Topic {topic_idx}:")
                print(", ".join([feature_names[i] for i in topic.argsort()[::-n_top_words - 1:-1]]))

        # Display top 10 words for each topic
        display_topics(H, feature_names, n_top_words=10)
```

Topic 0:

sleepy, slow, guy, beat, better, endorsed, debate, stage, look, rally

Topic 1:

vote, votes, opponent, lot, state, way, charge, asking, problem, forgiveness

Topic 2:

wins, mob, simple, anarchists, win, rioters, arsonists, flag, burners, look

Topic 3:

years, spent, damage, reversing, inflicted, horrible, period, year, country, true

Topic 4:

voted, tax, cuts, left, war, destroy, radical, american, agenda, ban